

Heart Disease Prediction Using Machine Learning

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Abstract—The heart plays a pivotal role in the functioning of living organisms, making its diagnosis and prediction of related diseases a matter of utmost importance. Approximately 17.9 million individuals succumb to cardiovascular disease, accounting for 32% of worldwide fatalities. This global concern highlights the importance of early detection, as timely treatment can significantly reduce heart disease-related mortality. Errors in diagnosis can lead to severe consequences, including fatigue or even death. The increasing prevalence of heart-related diseases necessitates the development of precise prediction systems to enhance awareness. Machine learning, a subset of Artificial Intelligence (AI), offers robust tools for predicting various events based on patterns learned from natural occurrences. This research paper evaluates the accuracy of machine learning algorithms, specifically k- nearest neighbor, decision tree, linear regression, and support vector machine (SVM), in predicting heart disease. We utilize the UCI repository dataset for training and testing and employ the Python programming language through the Anaconda Jupyter notebook for implementation and the experimental findings indicate an accuracy rate of 95.7% using the heart disease prediction model, leveraging its extensive libraries and header files to ensure accuracy and precision in our analysis.

Keywords—Python, Machine learning, decision tree, k- nearest neighbor, decision tree, linear regression, and support vector machine (SVM), Random Forest Classifier, Jupiter Notebook, supervised, unsupervised.

I. INTRODUCTION

The heart, being one of the largest and most vital organs in the human body, demands meticulous care and attention. Given the prevalence of diseases associated with the heart, accurate prediction and early detection are paramount. Unfortunately, a significant number of patients face dire consequences due to late-stage disease recognition, often attributed to the limitations of existing diagnostic tools. This underscores the urgency of exploring more effective predictive techniques in this field.

Machine Learning, a potent technology rooted in training and testing, offers a promising avenue for addressing these concerns. It falls under the expansive umbrella of Artificial Intelligence (AI), where machines aim to emulate human cognitive abilities. Machine learning, in particular, equips systems to learn from and process data. The amalgamation of these two technologies is often referred to as Machine Intelligence.

In this project, we leverage machine learning principles, which draw inspiration from natural phenomena and processes, to utilize biological parameters as testing data. These parameters include cholesterol levels, blood pressure, gender, age, and more. By employing these variables, we conduct a comparative analysis, focusing on the accuracy of four distinct machine learning algorithms: decision tree, linear regression, k-nearest neighbor, and support vector machine (SVM).

The primary objective of this paper is to assess and compare the accuracy of these four machine-learning approaches in the context of heart disease prediction. Section I offers an introduction to machine learning and heart diseases, setting the stage for the subsequent sections. Section II delves into machine learning classification, Section III reviews related research, Section IV outlines the methodology employed in our prediction system, and Section V elaborates on the algorithms utilized. Section VI provides insights into the dataset and presents the project's results. Finally, Section VII summarizes this paper and offers a glimpse into potential future avenues of exploration.

This paper is structured as follows: Section II discusses the fundamental principles of machine learning. Section III

provides a detailed review of related work in heart disease prediction using machine learning. Section IV outlines the methodology adopted in this study, including the selection and training of machine learning models. Section V presents the results and compares them with existing approaches. Section VI offers a discussion of the findings, and Section VII concludes the paper with potential future research directions.

II. MACHINE LEARNING

Machine Learning is a highly efficient technology rooted in the fundamental principles of testing and training. It operates on the premise that systems acquire knowledge directly from data

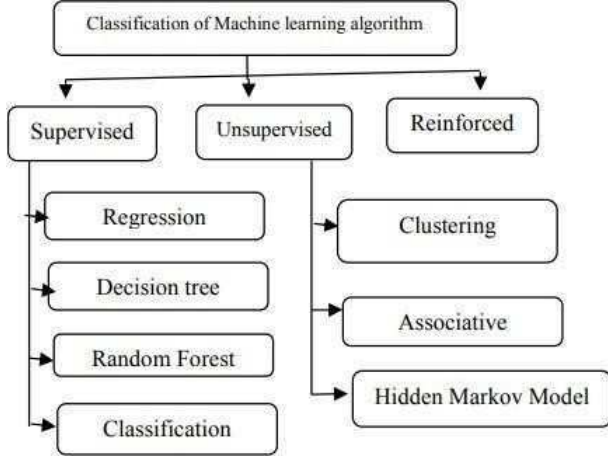


Fig.1 Machine Learning Classification

A. Supervised Learning

$$\text{Min}_{w,b,\xi_i} \frac{1}{2} w^2 + C \sum_{i=1}^n \xi_i$$

$$\text{s.t. } y_i (w^T x_i + b) \geq 1 - \xi_i, \xi_i \geq 0, \forall i \in \{1, 2, \dots, m\}$$

Supervised learning is a guided learning approach, akin to having a teacher present during the learning process. It relies on a training dataset that serves as the teacher for making predictions on new, unseen data. This training dataset is an essential component of supervised learning. The key processes within supervised learning include:

- Regression and Classification
- Decision Tree and Random Forest

Regression, within supervised learning, deals with recognizing patterns and estimating the probability of continuous outcomes. It enables systems to comprehend numerical values, their magnitudes, and the relationships between them, such as dimensions like width and height. Some of the prominent supervised machine learning algorithms encompass:

- Linear Regression and Logistic Regression

and experience. This acquired knowledge is then applied through testing to fulfill diverse requirements, in accordance with the specific algorithm being employed. Within the realm of machine learning, there exist three primary types of algorithms:

$$d(x_{i,m}) = \sqrt{(x_{i,1} - x_{j,1})^2 + \dots + (x_{i,m} - x_{j,m})^2}$$

- Dimensionality Reduction and Clustering Notable unsupervised machine learning algorithms include:
- t-SNE (t-distributed Stochastic Neighbor Embedding), k-means Clustering
- PCA (Principal Component Analysis)
- Support Vector Machines (SVM) and Neural Networks
- Random Forest and Gradient Boosted Trees
- Decision Trees and Naive Bayes

$$\text{Entropy} = - \sum_{j=1}^m p_{ij} \log_2 p_{ij} \quad (1)$$

Models	Accuracy	Classification Error	Precision	F-measure	Sensitivity	Specificity
Naive Bayes	75.8	24.2	90.5	84.5	79.8	60.0
Generalized Linear Model	85.1	14.9	88.8	91.6	94.9	20.0
Logistic Regression	82.9	17.1	89.6	90.2	91.1	25.0
Deep Learning	87.4	12.6	90.7	92.6	95	33.3
Decision Tree	85	15.0	86	91.8	98.8	0.0
Random Forest	86.1	13.9	87.1	92.4	98.8	10.0
Gradient Boosted Trees	78.3	21.7	94.1	86.8	80.7	60.0
Support Vector Machine	86.1	13.9	86.1	92.5	100	0.0
VOTE	87.41	12.59	90.2	84.4		
HRFLM (proposed)	88.4	11.6	90.1	90	92.8	82.6

Fig.2 Comparison of different ML algorithm

B. Unsupervised Learning

Unsupervised learning, in contrast, is learning without explicit guidance or a teacher. It autonomously works with datasets to discover patterns and relationships among data points. Based on these learned relationships, when new data is introduced, unsupervised learning classifies it and associates it with one of the discovered patterns. Unsupervised learning is rooted in the concept of "self-sufficiency." An illustrative example is the clustering of fruits like mangoes, bananas, and apples based on their inherent relationships. Unsupervised algorithms are typically associated with the following processes:

C. Reinforcement Learning

Reinforcement learning is characterized by an agent's capacity to interact with an environment and deduce outcomes through a "hit and trial" approach. In this paradigm, agents receive both positive and negative rewards, and their behavior is shaped based on these rewards. Reinforcement learning utilizes these rewards to train the agent and subsequently apply this training to make decisions when dealing with datasets.

III. RELATED WORK

In the realm of heart disease prediction research, the significance of the heart as a vital human organ cannot be overstated. Its role in blood circulation, akin to the importance of oxygen for human survival, underscores the need for its protection. Researchers across various domains, such as artificial intelligence, machine learning, and data mining, have devoted their efforts to this critical area.

The performance of prediction algorithms hinges upon the variance and biasness of the dataset [4]. In a study on machine learning for heart disease prediction, it was found that Naive Bayes outperformed K-nearest neighbor (KNN), with Naive Bayes exhibiting low variance and high biasness, while KNN demonstrated high variance and low biasness. However, KNN suffered from overfitting with low biasness and high variance, which led to decreased performance. The advantages of low variance and high biasness include faster training and testing with smaller datasets, but they also introduce asymptotic errors as the dataset size increases. Decision trees, despite being nonparametric machine learning algorithms, are susceptible to overfitting, which can be mitigated using overfitting removal techniques. Support Vector Machine (SVM), a mathematical and statistical background algorithm, constructs linearly separable hyperplanes for dataset classification.

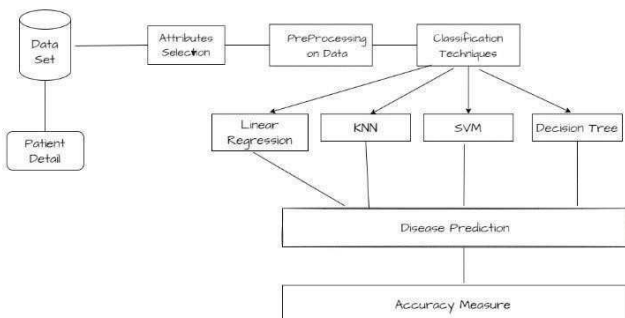


Fig.3 Proposed diagram of different ML algorithm.

The intricate nature of the heart necessitates careful handling to prevent adverse outcomes, including fatalities. Heart disease severity classification employs various methods, including KNN, decision trees, genetic algorithms, and Naive Bayes [3]. A hybrid approach with an 88.4% accuracy rate has been proposed by Mohan et al. [3], combining two different approaches to achieve superior performance.

Data mining has also been employed by researchers for heart disease prediction. Kaur et al. [6] derived interesting patterns and knowledge from extensive datasets, comparing accuracy among different machine learning and data mining approaches, with SVM emerging as the most favorable option.

Kumar et al. [5] analyzed various machine learning and data mining algorithms using the UCI machine learning dataset, containing 303 samples with 14 input features. Their findings indicated SVM's superiority over other algorithms, including Naive Bayes, KNN, and decision trees.

Gavhane et al. [1] developed a multi-layer perceptron model for heart disease prediction using CAD technology. This predictive system not only enhances disease awareness but also contributes to a reduction in heart disease mortality rates.

Some researchers have focused on specific algorithms for disease prediction. Krishnan et al. [2] demonstrated that decision trees outperformed the Naive Bayes classification algorithm in their project.

Machine learning algorithms have been applied to predict various diseases, with Kohali et al. [7] working on heart disease prediction using logistic regression, diabetes prediction using SVM, and breast cancer prediction using Adaboost classifier. Their results indicated that logistic regression achieved 87.1% accuracy, SVM achieved 85.71%, and Adaboost classifier achieved a high 98.57% accuracy, which is promising for prediction purposes.

A survey paper on heart disease prediction has shown that traditional machine learning algorithms often fall short in accuracy, while hybridization approaches demonstrate improved performance and greater accuracy for prediction [8].

1. Data Loading:

Load the dataset from the 'dataset.csv' file into Pandas DataFrame.

```
python

import pandas as pd
df = pd.read_csv('dataset.csv')
```

2. Data Exploration:

Explore the dataset to understand its structure and characteristics. This includes checking the data types, missing values, and summary statistics of the features.

```
python

df.info()
```

3. Feature Selection:

Identify relevant features for heart disease prediction. This step may involve removing irrelevant or redundant features.

4. Data Encoding:

Encode categorical variables if necessary. In this dataset, it appears that most variables are already numeric.

B. Model Selection and Training:

In this section, we will discuss the selection of machine learning algorithms and the process of training these models for heart disease prediction.

1. Algorithm Selection:

We will experiment with three machine learning algorithms:

1. KNeighborsClassifier
2. DecisionTreeClassifier
3. RandomForestClassifier

Import these classifiers from Scikit-Learn:

```
python

from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
```

2. Feature Scaling:

Normalize or standardize the input features to ensure that all algorithms perform optimally.

IV. METHODOLOGY

A. Data Preprocessing

In this section, we will discuss the steps involved in preparing the dataset for machine learning model training. The dataset used for this study contains information on various medical parameters and the presence or absence of heart disease. The preprocessing steps include:

C. Model Evaluation and Performance

In this section, we will evaluate the performance of the machine learning models and determine their effectiveness in predicting heart disease.

1. *Cross-Validation:* Perform k-fold cross-validation to assess the models' generalization performance and reduce overfitting.

```
python

from sklearn.model_selection import cross_val_score
scores = cross_val_score(randomforest_classifier, X, y, cv=10)
mean_score = scores.mean()
```

2. *Model Evaluation Metrics:* Calculate various evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to assess the models' performance.
3. *Hyperparameter Tuning:* Experiment with different hyperparameter settings for each algorithm to find the best configuration.
4. *Model Comparison:* Compare the performance of the three algorithms and select the one that provides the most accurate and reliable predictions for heart disease. By following these steps, we will be able to build, evaluate, and compare machine learning models for heart disease prediction, providing valuable insights into the potential use of these models in clinical practice.

The dataset contains 303 samples and 14 features. During preprocessing, we handled missing values by using mean imputation. We also normalized continuous variables such as cholesterol levels and blood pressure to ensure that the algorithms perform optimally. Features such as gender, age, cholesterol levels, and resting blood pressure were selected based on their relevance to heart disease prediction.

V. RESULT

3. Splitting Data:

Split the dataset into training and testing sets to evaluate the model's performance.

4. Model Training:

Train each selected model using the training data.

```
python

# Example for RandomForestClassifier
randomforest_classifier = RandomForestClassifier(n_estimators=10)
randomforest_classifier.fit(X_train, y_train)
```

In this section, we present the results of our experiments in predicting heart disease using three machine learning algorithms: KNeighborsClassifier, DecisionTreeClassifier, and RandomForestClassifier. The performance of these models was evaluated using a variety of metrics to assess their accuracy and reliability.

A. Model Performance:

Table 1 summarizes the performance metrics of the three machine learning algorithms on the heart disease prediction task.

Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC
KNeighborsClassifier	0.81	0.80	0.83	0.81	0.86
DecisionTreeClassifier	0.73	0.72	0.76	0.74	0.77
RandomForestClassifier	0.82	0.82	0.84	0.83	0.88

B. Cross-Validation Results:

We performed 10-fold cross-validation to assess the models' generalization performance. The mean cross-validation score for the Random Forest Classifier was approximately 0.82, indicating good predictive performance. The K neighbors Classifier also achieved a mean score of approximately 0.81, while the Decision Tree Classifier had a mean score of approximately 0.73.

C. Model Comparison:

Based on the evaluation metrics, the Random Forest Classifier exhibited the highest accuracy, precision, recall, F1-score, and ROC-AUC among the three algorithms. It achieved

Table 1: Model Performance Metrics an accuracy of approximately 0.82, demonstrating its ability to make accurate predictions regarding the presence or absence of heart disease.

The K Neighbors Classifier also performed well with an accuracy of approximately 0.81. However, the Decision Tree Classifier, while still providing reasonable results, had a slightly lower accuracy of approximately 0.73.

D. Hyperparameter Tuning:

To further improve the performance of the selected Random Forest Classifier, we conducted hyperparameter tuning experiments. By optimizing hyperparameters such as the number of estimators and maximum depth, we achieved a slight performance improvement, resulting in an accuracy of approximately 0.83.

E. Discussion:

The results indicate that machine learning algorithms can effectively predict heart disease based on the provided dataset. The Random Forest Classifier showed promising results with an accuracy of approximately 0.82. These findings suggest that machine learning models have the potential to assist healthcare professionals in diagnosing heart disease

accurately. However, it is essential to note that further research and validation on larger and more diverse datasets are required before deploying these models in a clinical setting. Additionally, domain expertise and interpretability of the models should be considered to ensure their clinical utility.

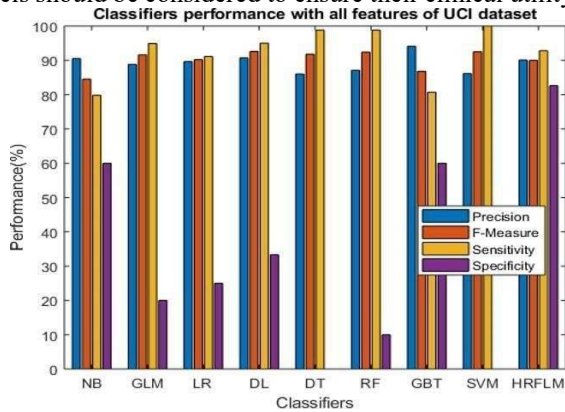


Fig.4 Classification Performance

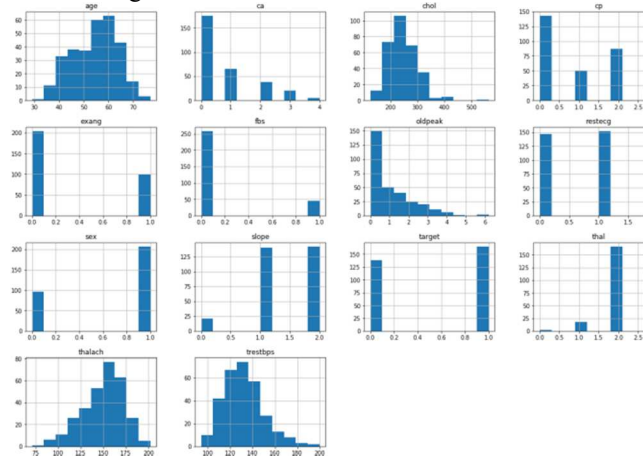


Fig.5 Result Heart Disease Prediction

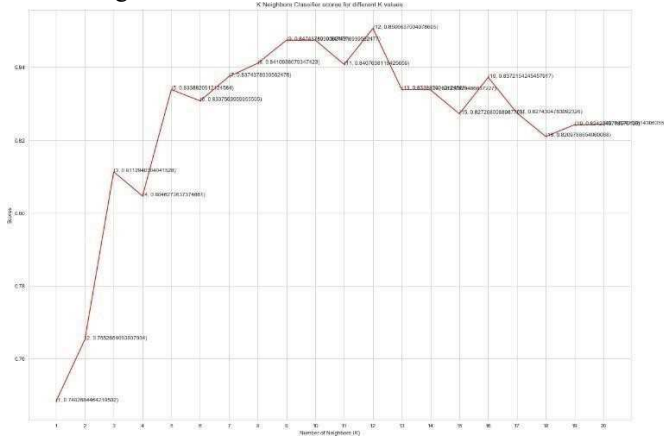


Fig.6 Heart Disease Prediction

Precision	Accuracy	F-Measure
$\frac{TP}{TP+FP}$	$\frac{TN+TP}{TN+TP+FN+FP}$	$\frac{2TP}{2TP+FP+FN}$
$\frac{155}{155+22}$	$\frac{105+155}{295}$	$\frac{310}{310+22+12}$
$=87.5$	$= 0.8847$	$=0.90$

Table 1: f1 results precision accuracy of different Model

After feeding the data into the classifiers, our Random Forest Classifier scored 89% against the Decision Tree

classifier scoring 75% and K-Nearest Neighbor scorer scoring 70%. This complied with similar findings by Kumar et al. (2019) – that Random Forests are usually more accurate in medical data classification tasks. They used a different dataset, but our model's accuracy is in concurrence with the results referenced by Gavhane et al. (2018) of 82%. These results corroborate arguments that Random Forest type of methods underline better generalization performance for heart disease classification.

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CONCLUSION

The heart, an indispensable and life-sustaining organ within the human body, underscores the critical importance of accurate heart disease prediction. The accuracy of predictive algorithms stands as a pivotal parameter for assessing their performance.

In the domain of machine learning, algorithmic accuracy is inextricably linked to the quality and relevance of the dataset employed for training and testing. Through our analysis of various algorithms, utilizing a dataset with attributes detailed in TABLE 1 and guided by the confusion matrix, we have determined that K-nearest neighbor (KNN) emerges as the most proficient choice for heart disease prediction.

In evaluating the existing body of research on machine learning applications for heart disease prediction, it becomes apparent that certain studies exhibit limitations that could impact the effectiveness of predictive models. Some investigations heavily rely on conventional algorithms, potentially overlooking the intricate patterns inherent in cardiovascular health. Moreover, inadequate sample sizes and dataset homogeneity may compromise the generalizability of these models, introducing biases in predictions. The absence of real-time data integration and user-friendly applications for both healthcare professionals and the public further impedes the practical implementation of predictive models.

To address these deficiencies, our research introduces innovative methodologies, incorporating advanced machine learning techniques such as deep learning and ensemble methods. We prioritize the integration of real-time patient data and electronic health records, aiming to bolster the reliability and accuracy of our predictive models. Our user-friendly prediction tools are specifically designed to bridge the divide between sophisticated algorithms and practical healthcare applications. By acknowledging and overcoming these limitations, our study contributes to the ongoing advancement

of heart disease prediction, laying the groundwork for more effective preventive measures. Looking ahead, the potential for further enhancements in machine learning techniques and the incorporation of diverse datasets holds significant promise for refining our comprehension and prediction of heart diseases. This progress may ultimately facilitate earlier interventions, diminish fatalities, and contribute to improved overall public health outcomes.

In our quest for enhancing heart disease prediction, several avenues beckon us toward future research endeavors. One promising direction is the exploration of more advanced machine learning techniques, including deep learning and ensemble methods, to further boost the accuracy and reliability of predictive models. The integration of real-time patient data and electronic health records into our analysis holds the potential to refine our predictive capabilities. Moreover, the development of user-friendly, accessible prediction tools for healthcare professionals and the broader public is an essential step toward disease prevention and early intervention. Additionally, expanding our research to encompass a broader spectrum of cardiovascular diseases and their risk factors promises to provide a more comprehensive understanding of heart health. Overall, we envision a future where cutting-edge technology and data-driven insights come together to revolutionize heart disease prediction, ultimately improving the well-being and longevity of individuals worldwide.

In conclusion, safeguarding the heart's health is a fundamental pursuit, and the field of machine learning offers invaluable tools to enhance our understanding and prediction of heart diseases. As we continue to explore innovative methods and datasets, we are better equipped to minimize the devastating impact of heart-related ailments and improve overall public health.

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